

THE INSIDR

Developer Specification Document

Complete Platform Specification — Confirmed & Finalised

Version 2.0 | March 2026 | Confidential

Detail	Information
Platform Name	The Insidr
Platform Type	AI Trading Intelligence — Web Application
Target Market	Retail traders — Forex, Stocks, Indices, Crypto
Subscription Tiers	Pro (\$28/mo) and Elite (\$79/mo)
Total Pages	13 pages including landing page and onboarding
Document Status	Finalised — approved for development

SECTION 1: EXECUTIVE SUMMARY

Executive Summary

What The Insidr Is

The Insidr is a web-based AI-powered trading intelligence platform for retail traders. It does not execute trades. It provides research, analysis, signals, and performance tools that help retail traders make better informed decisions across Forex, Stocks, Indices, and Cryptocurrency markets. Every feature is designed to close the gap between how retail traders and institutional traders approach the market.

Mission Statement

Mission

To give retail traders access to the same institutional-grade market intelligence, analysis, and decision-making tools that professional traders use — delivered through a single platform at a price accessible to everyone.

Subscription Model

Tier	Monthly	Annual	Annual Saving
Pro	\$28/month	\$280/year	\$56 (2 months free)
Elite	\$79/month	\$790/year	\$158 (2 months free)

Complete Platform Content

AI Trade Ideas

Real-time trade setups generated by a multi-layer confluence engine combining technical price action analysis and fundamental macro drivers. Ideas are published only when a genuine high-conviction setup is identified. Every idea includes entry zone, stop loss, TP1/TP2/TP3, R:R ratios, confidence score (0-100%), technical reasoning, fundamental reasoning, key invalidation level, risk environment alignment badge, and a full idea status tracker.

Three timeframe classifications:

- Scalp — uses 1m and 5m chart timeframes
- Day Trade — uses 5m, 15m, and 1H chart timeframes
- Swing — uses 1H, 4H, and Daily chart timeframes

Technical signals used in idea generation:

- Market structure — Break of Structure (BOS), Change of Character (CHoCH), equal highs and lows
- Key price levels — support/resistance zones, supply and demand zones, psychological round numbers
- Key liquidity levels — previous day high/low, previous week high/low, previous month high/low, equal highs/lows as liquidity pools
- Volume profile — Point of Control (POC), Value Area High (VAH), Value Area Low (VAL)
- VWAP — session VWAP and weekly VWAP with plus and minus 1 standard deviation bands
- 200 EMA — on 4H and Daily timeframes only
- Trend lines — auto-detected from pivot points, minimum 2 touches to validate
- Chart patterns — 13 types including Head and Shoulders, double tops/bottoms, flags, wedges, triangles
- Candlestick signals — engulfing candles, pin bars, doji, inside bars at significant price levels only

Fundamental signals used in idea generation:

- Yield differential per pair — 2yr and 10yr spread direction and rate of change
- Central bank policy divergence
- Fundamental driver profiles per asset — hardcoded per pair
- Live economic data — CPI, GDP, PMI, Employment
- AI-scored news sentiment
- Cross-market correlations — Oil/CAD, JPY/equities, Gold/yields, AUD/commodities etc
- Risk On/Off environment — 7 macro indicators

Technical Chart Analysis

Pure price action chart analysis generated by AI for every asset on a user's watchlist. Uses the same technical signals as the trade idea engine. No lagging indicators. Multi-timeframe analysis panel covers 1H, 4H, and Daily simultaneously. Analysis appears on the Trade Idea Detail page showing the specific setup with key levels marked on the chart. Also includes what supports the current bias and what could flip it with specific price levels referenced throughout.

AI Bias Cards

Swing trade bias and day trade bias generated per asset every 30 minutes. Each card shows directional label, strength qualifier, what supports the bias (4 bullet points), and what could flip it (4 bullet points with specific price levels).

Live News and Sentiment

Real-time financial news feed covering all four markets with AI sentiment scoring on every article. Market sentiment gauges (Forex, Stocks, Indices, Crypto) updated every 30 minutes with score 0-100, directional label, and driver summary. Most bullish and most bearish assets ranked in real time. Sentiment is calculated from news sentiment (40%), price momentum (40%), and volume/volatility signals (20%).

Economic Calendar

Full economic calendar with AI pre-event analysis predicting likely market impact across three scenarios (above/below/in-line with forecast) and AI post-event analysis published within minutes of each data release. Daily and weekly views. Covers all major currencies with impact filtering, watchlist filtering, countdown timers, and per-event View Insidr Data button.

Economy Pages

Dedicated macro intelligence pages for nine countries: USA, Eurozone, UK, Japan, Australia, New Zealand, Canada, Switzerland, and China. Each page covers central bank rate and meeting expectations, 2yr and 10yr yield levels, yield rate of change chart, CPI/GDP/PMI/Employment data, risk sentiment implication for that currency, recent high-impact news, and a View Insidr Data button per individual item.

Yield Differential Engine

2yr and 10yr government bond yield spreads calculated per currency pair. Shows which country holds the yield advantage, whether the spread is widening or narrowing, and whether the pair is aligned with or diverging from yield direction. Yield rate of change charts on the asset breakdown show both countries simultaneously with Levels/Spread/RoC toggle buttons and 1W/1M/3M period selector.

Risk On / Risk Off Engine

Global macro sentiment scored every 15 minutes across seven indicators: VIX, S&P 500, JPY strength, Gold direction, US 10-year yield, DXY, and commodity index. Outputs Risk On, Neutral, or Risk Off. Boosts confidence on ideas aligned with the environment and suppresses contradicting ideas. Displayed on the dashboard widget and on every trade idea card.

Watchlist

User-configured asset monitor grouped by market with live prices, 24hr change, and AI bias badge per row. Clicking any asset loads it on the Dashboard as the selected asset breakdown. Collapsible market groups. Add asset via search modal.

AI Chat Assistant

Persistent AI trading mentor available on every page via floating widget. Remembers conversation history across sessions. Has full context of the user's watchlist, preferred markets, experience level, win rate, and trading performance. Can analyse assets on demand, review trades, explain events, and educate users. Pro users: 50 messages per day. Elite users: unlimited.

Trade Journal and Portfolio Tracker

Tradezella-style trading journal with four tabs: Overview, Journal, Analytics, and Calendar view. Users log trades manually, import via CSV, or sync automatically via MT4/MT5 Expert Advisor, IBKR Flex API, Binance API, or Coinbase API. Full analytics including win rate, R:R, profit factor, drawdown, performance by market/session/day/strategy/emotional state/mistake tag, equity curve, and mistake cost tracker.

Alerts and Notifications

Five alert types: trade idea alerts, economic calendar alerts, news alerts, risk sentiment shift alerts, and bias change alerts. Two delivery channels at launch: in-app notification centre and email. Every alert type has independent on/off toggles per channel. SMS to be added in a future release.

View Insider Data

A universal magnifying glass icon button that appears next to every individual item across the platform — news articles, calendar events, trade ideas, assets, economy page components. Generates a focused Claude-powered AI briefing about that specific item only. Never a page-level summary — always item-level intelligence. Cached 15-30 minutes per item.

Key Differentiators

- Trade ideas generated from a genuine multi-layer confluence of technical price action and fundamental macro analysis
- Risk On/Off engine actively suppresses ideas that contradict the global macro environment
- Institutional-grade yield differential and rate of change analysis surfaced at the individual pair level
- Tradezella-style journal combined with forward-looking trade ideas, news, and AI signals in one subscription

- AI chat assistant with persistent memory personalised to each user's trading history
- Economy pages giving traders a complete country-level macro picture for all major currencies
- Pure price action analysis — no lagging indicators, no RSI, no MACD, no Bollinger Bands, no Fibonacci
- View Insider Data button providing contextual AI intelligence at the individual item level across the entire platform

SECTION 2: DESIGN SYSTEM

Design System

Colour Palette

Variable	Hex Value	Usage
--void	#080810	Main app background — deepest black
--base	#0d0d1a	Sidebar, secondary backgrounds
--card	#11111f	Card backgrounds
--surface	#1a1a2e	Elevated surfaces, inputs
--raised	#1e1e35	Hover states
--purple	#7b2fff	Primary accent — bearish signals, AI features, active states
--purple-bright	#9d5cff	Purple highlights and text
--purple-subtle	rgba(123,47,255,0.08)	Purple tinted backgrounds
--gold	#d4900a	Secondary accent — bullish signals, premium features
--gold-bright	#f5b533	Gold highlights and text
--gold-subtle	rgba(212,144,10,0.08)	Gold tinted backgrounds
--green	#00d68f	Positive P&L, bullish, profit, open/active states
--red	#ff4757	Negative, bearish, loss, high impact events
--text-primary	#f0f0ff	Primary text
--text-secondary	#8888aa	Secondary text
--text-muted	#44445a	Muted and disabled text
--border	rgba(255,255,255,0.06)	Default borders
--border-purple	rgba(123,47,255,0.3)	Purple accent borders
--border-gold	rgba(212,144,10,0.3)	Gold accent borders

Typography

Font	Usage	Google Fonts Import
Bebas Neue	All large headings, section titles, asset names, large numbers, P&L figures. Always use letter-spacing: 2-3px.	Yes — free
DM Sans	All body text, labels, descriptions, nav items, buttons. Weights: 300, 400, 500, 600, 700.	Yes — free
JetBrains Mono	All prices, percentages, yield values, timestamps, numerical API data. Weights: 400, 500, 600.	Yes — free

Component Standards

Border Radius

Standard elements: 8px. Larger containers and modals: 12px. Status dots and avatars: 50% (full circle). Never fully square or fully pill-shaped for primary UI elements.

Theme

Dark theme only. No light mode. Every background, surface, card, and input uses the dark colour variables above. Body background is always #080810.

Charts

TradingView Advanced Chart Widget for all candlestick charts — free embed, no API key required. Recharts library for all data visualisation charts (yield RoC, journal analytics, equity curves). Bearish candles use purple (#7b2ff). Bullish candles use green (#00d68f).

SECTION 3: GLOBAL ARCHITECTURE

Global Architecture

Navigation — Left Sidebar

Fixed left sidebar (220px width) present on every authenticated page. Collapsible to icon-only mode. Not present on Landing Page, Auth pages, or Onboarding.

Navigation items:

- Dashboard
- Trade Ideas
- News and Sentiment
- Economic Calendar
- Economy
- Watchlist
- Journal
- Alerts
- Settings

Bottom of sidebar: User pill showing avatar initials, name, plan badge (PRO/ELITE), and chevron for account menu.

Top Bar

Horizontal top bar (56px height) on every authenticated page. Contains the pairs watchlist tab strip with live prices and 24hr percentage change, a plus button to add pairs, notification bell icon with unread count badge, and profile icon. Sticky positioned.

Authentication Flow

Flow	Route	Details
Sign Up	/auth/signup	Email + password + name. Supabase Auth. After signup goes to onboarding.
Sign In	/auth/signin	Email + password. Supabase Auth. After signin goes to

Flow	Route	Details
		dashboard.
Forgot Password	/auth/forgot	Email input. Supabase password reset email.
Reset Password	/auth/reset	New password + confirm. Supabase reset link handler.

Subscription Gating

Every authenticated page checks Supabase for an active subscription record. No active subscription redirects to /pricing. Pro tier users have AI chat limited to 50 messages per day. Elite tier users have unlimited access to all features. Middleware reads from the subscriptions table on every page load.

Complete Page Routes

Page	Route	Access
Landing Page	/	Public
Sign Up	/auth/signup	Public
Sign In	/auth/signin	Public
Forgot Password	/auth/forgot	Public
Pricing	/pricing	Public
Onboarding	/onboarding	Auth required — runs once post-signup
Dashboard	/dashboard	Auth + subscription
Trade Ideas	/ideas	Auth + subscription
Trade Idea Detail	/ideas/[id]	Auth + subscription
News and Sentiment	/news	Auth + subscription
Economic Calendar	/calendar	Auth + subscription
Economy Landing	/economy	Auth + subscription
Economy Country	/economy/[country]	Auth + subscription
Watchlist	/watchlist	Auth + subscription
Journal	/journal	Auth + subscription
Alerts	/alerts	Auth + subscription
Settings	/settings	Auth required

SECTION 4: ONBOARDING FLOW

Onboarding Flow

Full-screen wizard that runs once after signup. Users cannot access the platform until onboarding is complete. Progress bar at top showing current step. The Insidr logo centered. Back and Next buttons on every step. Background analytics track every step entry, completion, time spent, and drop-off point.

Step 1 — Experience Level

- Heading: HOW EXPERIENCED ARE YOU?
- Three large clickable cards: Beginner (0-1 years) / Intermediate (1-3 years) / Advanced (3+ years)
- Single select. Personalises AI chat assistant tone and explanation depth throughout the platform.

Step 2 — Preferred Markets

- Heading: WHAT DO YOU TRADE?
- Four large clickable cards: Forex / Stocks / Indices / Crypto
- Multi-select. Personalises dashboard, news feed, trade ideas, and sentiment to selected markets.

Step 3 — Account Currency

- Heading: WHAT IS YOUR ACCOUNT CURRENCY?
- Eight currency options: USD, GBP, EUR, AUD, CAD, JPY, NZD, CHF
- Single select. Used for display across the platform.

Step 4 — Build Watchlist

- Heading: ADD YOUR FIRST ASSETS
- Pre-populated asset suggestions based on markets selected in Step 2
- Search input to find any asset

- Toggleable asset chips — click to add or remove
- Minimum 1 asset required to proceed

Step 5 — Choose Subscription Plan

- Heading: CHOOSE YOUR PLAN
- Pro vs Elite side-by-side comparison with full feature list
- Monthly/Annual billing toggle — annual saves 2 months
- Pro: \$28/mo or \$280/yr. Elite: \$79/mo or \$790/yr.
- 7-day free trial on both plans
- Stripe Checkout on plan selection

Completion

- Brief success animation
- YOUR DASHBOARD IS READY heading in Bebas Neue
- Start Trading gold button redirects to /dashboard

Funnel Analytics

Every onboarding step fires an analytics event capturing: step name, user ID, timestamp, time spent on step, whether the user proceeded or went back, and whether they abandoned. This data is used to identify friction points and optimise the flow over time. Recommended tool: PostHog or Mixpanel.

Event Name	Fires When
onboarding_step_viewed	User lands on any step
onboarding_step_completed	User clicks Next on any step
onboarding_step_back	User clicks Back
onboarding_abandoned	User closes or navigates away mid-flow
onboarding_completed	User reaches the completion screen
plan_selected	User selects Pro or Elite
trial_started	Stripe trial confirmed

SECTION 5: DASHBOARD

Dashboard

The Dashboard is the main hub of the platform. It serves as both the user's daily market briefing and the primary asset analysis view. When a user selects any asset from the pairs tab strip or watchlist, the Dashboard loads the full asset breakdown for that asset in the center area.

Global Layout

- Left sidebar — collapsible to icon-only mode, contains all navigation
- Top bar — pairs tab strip with live prices, notification bell, profile icon
- Main content area — asset breakdown center, news sidebar right (collapsible)

Dashboard Components

1. AI Daily Briefing Bar

Full-width bar below the top bar. Purple-to-gold subtle gradient background. Contains a small gradient icon and a Claude-generated personalised briefing covering key overnight price moves, what to watch today, and upcoming risk events. Generated once per morning per user, cached daily, personalised to their watchlist and preferred markets.

2. Session Clock

Four trading session indicators: London, New York, Tokyo, Sydney. Each shows a status dot (green for open, gold for overlap, grey for closed), session name, status tag, and time range. Open and overlapping sessions show an animated pulse dot. Overlap periods (London/New York) are highlighted in gold as the highest volatility windows.

3. Pairs Tab Strip

Horizontally scrollable strip in the top bar showing the user's watchlist assets. Each tab shows asset symbol, live price (updating via WebSocket), and 24hr percentage change (green for positive, red for negative). Clicking a tab loads that asset in the main chart area. A plus button opens the add asset modal.

4. TradingView Chart

The main centerpiece of the dashboard. TradingView Advanced Chart Widget embedded via script tag — free, no API key. Symbol automatically syncs to the selected pair tab. Users can switch timeframes, use drawing tools, and expand to full screen.

5. AI Bias Cards (overlaid on chart)

Three cards overlaid top-left of the TradingView chart with frosted glass treatment (backdrop-filter: blur(8px)): Swing Trading bias, Day Trading bias, and News Impact level. Each shows directional label and strength qualifier. Generated every 30 minutes and cached. Clicking any bias card shows the full supports/flips breakdown.

6. Live Price Overlay (on chart)

Frosted glass panel overlaid top-right of the chart showing the current live price, daily high, daily low, and 24hr change percentage. Updates in real time via WebSocket.

7. What Supports the Bias

Panel below the chart left side. Four bullet points with purple dots listing specific technical and fundamental reasons the current bias is valid. References actual price levels throughout. Generated by Claude at same time as bias cards.

8. What Could Flip It

Panel below the chart right side. Four bullet points with gold dots listing specific price levels and events that would invalidate the current bias. Always references exact price levels. Generated alongside the bias cards.

9. View Insider Data Button

Full-width button below the supports/flips panels. On click generates a comprehensive Claude-powered briefing covering everything about the selected asset: current bias, yield direction and RoC, Risk On/Off alignment, key price levels (POC, VWAP, 200 EMA, S/R), recent news sentiment, fundamental context, and upcoming calendar events. Cached 15-30 minutes per asset. Expands as a readable panel below the button.

10. Live News Sidebar (right, collapsible)

Fixed right sidebar showing a live news feed filtered to the selected asset. Simplified card format: sentiment badge (Bullish/Bearish/Neutral) + headline + timestamp. High and Medium impact only. Auto-refreshes every 60 seconds. New articles slide in at top. Collapsible to give full width to the chart area. Each article has a View Insider Data button.

11. Risk On/Off Widget

Displays the current global risk environment label (Risk On / Neutral / Risk Off) with individual votes for all 7 indicators: VIX, S&P 500, JPY, Gold, US 10Y Yield, DXY, Commodity Index. Each indicator shows its current reading and its vote. Recalculated every 15 minutes.

Asset Breakdown — Below Chart Sections

The full asset breakdown below the chart contains: What Supports the Bias, What Could Flip It, View Insider Data button, and then a two-column section:

- Left column — Recent News Feed: simplified news cards (sentiment badge + headline + timestamp) filtered to the selected asset, High and Medium impact only, newest first, scrollable, auto-refreshes every 60 seconds. Each article has View Insider Data button.
- Right column — Yield Line Chart: Recharts LineChart showing 2yr and 10yr yield data for the countries relevant to the selected pair. For EUR/USD this is German 2yr/10yr vs US 2yr/10yr (4 lines). Three toggle buttons: Levels (actual % values) / Spread (differential) / RoC (daily rate of change %). Period selector: 1W / 1M / 3M. Zero line on RoC view. Green shading above zero, red below.

SECTION 6: TRADE IDEAS

Trade Ideas

Trade Ideas is the core value proposition of The Insidr. AI-generated trade setups published in real time only when a genuine high-conviction setup is identified through multi-layer confluence scoring. Quality over quantity — ideas are never generated on a schedule.

Trade Ideas Page (/ideas)

- Page header: Trade Ideas title + live count of active ideas with green dot
- Platform performance stats bar: overall win rate, average R:R, TP1/TP2/TP3 hit rates displayed publicly to build trust
- Filter bar: Market (All/Forex/Stocks/Indices/Crypto), Timeframe (All/Scalp/Day Trade/Swing), Confidence (All/High 80%+/Medium/Low), Direction (All/Long/Short), Sort (Newest/Highest Confidence/Best R:R)
- Three-column card grid on desktop

Trade Idea Card

- Direction badge (LONG in gold / SHORT in purple)
- Timeframe badge (SCALP / DAY TRADE / SWING)
- Confidence badge with emoji (High = fire, Medium = lightning, Low = bulb) + percentage
- Asset name + full name
- Time posted
- Entry zone, stop loss, TP1/TP2/TP3 with R:R per target
- Two-line reasoning preview
- Status indicator (Active / Entry Hit / TP1 Hit / TP2 Hit / TP3 Hit / Stopped / Invalidated)
- View Full Idea button + Set Alert button
- Card top border: 2px gradient — purple for Short, gold for Long

Trade Idea Detail Page (/ideas/[id])

- Direction, timeframe, and confidence badges
- Asset name, time generated, status tracker (Generated > Entry Hit > TP1 > TP2 > TP3 > Closed with checkmarks)

- TradingView chart pre-loaded with asset, entry/SL/TP levels as horizontal lines
- Trade levels panel: entry zone, SL, TP1/TP2/TP3 with R:R ratios
- Technical reasoning paragraph
- Fundamental reasoning paragraph
- Key invalidation level and description
- Risk environment alignment badge
- View Insider Data button — explains why this specific idea was generated
- Action bar: Set Alert / Add to Journal / Ask AI
- Legal disclaimer: This trade idea is generated by AI for informational purposes only. It does not constitute financial advice. Always conduct your own research and never risk more than you can afford to lose.

Timeframe Classifications

Type	Chart Timeframes Used	Typical Hold Time
Scalp	1m, 5m	Minutes to a few hours
Day Trade	5m, 15m, 1H	Intraday — opened and closed same day
Swing	1H, 4H, Daily	2 to 7 days

Idea Lifecycle

Status	Description
Active	Setup identified, awaiting price to enter the entry zone
Entry Hit	Price entered the entry zone — trade is live
TP1 Hit	First take profit target reached
TP2 Hit	Second take profit target reached
TP3 Hit	Third take profit target reached
Stopped Out	Stop loss level was hit
Invalidated	Invalidation level was breached before entry
Expired	Idea reached its maximum age without triggering — Scalp: 24hrs, Day Trade: 24hrs, Swing: 7 days

Generation Engine — Technical Signals

- Market structure — BOS, CHoCH, equal highs and lows as liquidity pools
- Key price levels — support/resistance zones, supply and demand zones, psychological round numbers

- Key liquidity levels — previous day high/low, previous week high/low, previous month high/low
- Volume profile — POC, VAH, VAL
- VWAP — session and weekly with plus/minus 1 SD bands
- 200 EMA — 4H and Daily only
- Trend lines — minimum 2 pivot point touches to validate
- Chart patterns — 13 types: H&S, inverse H&S, double top/bottom, triple top/bottom, bull/bear flags, ascending/descending/symmetrical triangles, rising/falling wedges
- Candlestick signals — engulfing candles, pin bars, hammers, shooting stars, doji, inside bars — only flagged at significant price levels

Generation Engine — Fundamental Signals

- Yield differential — 2yr and 10yr spread direction and rate of change per pair
- Central bank policy divergence
- Fundamental driver profiles — hardcoded per asset covering key economic drivers, sentiment indicators, and typical behaviour
- Live economic data — CPI, GDP, PMI, Employment readings
- AI-scored news sentiment from last 2 hours
- Cross-market correlations — AUD/NZD with commodities and China, CAD with oil, CHF with risk sentiment, JPY with equities, Gold with USD and yields
- Risk On/Off environment — 7 indicator composite score

Confluence Scoring

Every potential setup is scored across all technical and fundamental layers. The total confluence score determines the confidence percentage and whether the idea is published. Minimum threshold to publish: 50% confidence. Ideas are suppressed below this threshold regardless of how strong individual signals appear.

Risk environment alignment adds +2 confluence points when idea direction aligns with the current Risk On/Off state and deducts -3 points when it contradicts, potentially suppressing the idea entirely.

Yield differential alignment adds up to +3 points when both 2yr and 10yr spreads confirm the idea direction and deducts up to -3 points when they contradict.

SECTION 7: NEWS & SENTIMENT

News & Sentiment

Layout

Split layout. Left side (approximately 65% width): scrolling news feed. Right side (approximately 35% width, sticky): sentiment panel with market gauges and bullish/bearish asset rankings.

Filter Bar

- Market tabs: All / Forex / Stocks / Indices / Crypto
- Impact toggle: All / High / Medium / Low
- Asset search: type any symbol to filter news to that specific asset
- Active filters displayed as removable tags

News Feed

- Chronological, newest first
- Auto-updates every 60 seconds — new articles slide in at top
- Each card: impact badge (HIGH/MED/LOW) + sentiment badge (Bullish/Bearish/Neutral) + headline (bold) + 2-line AI summary + timestamp + source + asset tags with direction arrows
- Expand toggle: shows full AI analysis paragraph + link to original source + Ask AI button
- View Insidr Data button on every article — explains what that specific article means for affected assets

Sentiment Panel (Right)

- 4 market sentiment gauges: Forex / Stocks / Indices / Crypto
- Each gauge: score 0-100, directional label (Strongly Bearish through Strongly Bullish), progress arc, 1-line driver text
- Most Bullish Assets: top 5 ranked by combined sentiment score, clicking filters the news feed to that asset
- Most Bearish Assets: top 5 ranked, same interaction

Sentiment Calculation

Composite score calculated every 30 minutes per market and per asset. Three weighted components:

- News sentiment (40%) — average Claude API sentiment score of last 20 articles for the market/asset
- Price momentum (40%) — 24hr percentage price change of major assets within that market
- Volume/volatility signals (20%) — unusual volume signals from Twelve Data

Score range 0-100. Labels: 0-20 Strongly Bearish, 21-40 Bearish, 41-59 Neutral, 60-79 Bullish, 80-100 Strongly Bullish. Cached in Redis with 30-minute TTL.

Important Distinction

The 4 market sentiment gauges are different from the Risk On/Off engine. Sentiment gauges measure the mood within a specific market (Forex/Stocks/Indices/Crypto) based on news and price. Risk On/Off measures global macro risk appetite across 7 specific instruments. They can conflict — e.g. Crypto sentiment Bullish while Risk Off is active globally.

SECTION 8: ECONOMIC CALENDAR

Economic Calendar

Views

- Daily View (default) — today's events grouped into Upcoming and Released sections
- Weekly View — 7-day grid, colour coded by impact level (red=high, gold=medium, grey=low), today's column highlighted

Filter Bar

- Impact: All / High / Medium / Low
- Currency multiselect with flags: USD, EUR, GBP, JPY, AUD, CAD, NZD, CHF
- Market relevance: Forex / Stocks / Indices / Crypto
- Watchlist toggle: show only events affecting user's watched assets

Per Event Display

- Event name
- Date and time in user's local timezone
- Impact badge (High/Medium/Low)
- Country flag + currency code
- Previous value
- Forecast value
- Actual value (shown when released)
- Beat/Miss/Inline indicator after release (green checkmark for Beat, red X for Miss, grey arrow for Inline)
- Live countdown timer for upcoming events
- View Insidr Data button on every event

Event Detail Panel (on click)

- What is this? — plain English explanation of the economic indicator

- Expected Market Impact — three scenario bullets: above forecast, in-line, below forecast with specific asset reactions for each
- Most Affected Assets — ranked list of asset tags
- Set Alert button

Post-Event Analysis

Generated within minutes of each actual figure being published. The backend monitors Twelve Data for actual value updates, polling every 60 seconds around scheduled event times. When an actual value is detected, a post-event job fetches 15-minute price changes for all affected assets and sends everything to Claude for analysis. The generated analysis covers beat/miss assessment, fundamental implications, and actual price reactions of major affected assets.

SECTION 9: ECONOMY PAGES

Economy Pages

The Economy section provides complete macro intelligence for nine countries. Each country has a dedicated page. A country selector tab strip at the top allows instant switching between all nine countries. Accessible from the left navigation bar.

Nine Countries

USA, Eurozone (Germany as proxy), United Kingdom, Japan, Australia, New Zealand, Canada, Switzerland, China.

Economy Landing Page (/economy)

Grid of nine country cards each showing the country flag, name, currency code, central bank name, and current interest rate. Clicking any card navigates to that country's detail page.

Country Page Structure

1. Country Header

Flag, country name, central bank name, currency code, last data updated timestamp.

2. Central Bank Panel

Current interest rate, rate direction (Hiking/Cutting/Holding), overall cycle trend, next meeting date with live countdown, market-implied probability of cut/hold/hike at next meeting. View Insider Data button explains the current central bank stance and its implications for that currency.

3. Yield Levels Panel

Current 2yr yield with daily basis point change. Current 10yr yield with daily basis point change. Yield curve status (Normal/Flat/Inverted) calculated from the 2s10s spread. Inverted curve highlighted in red as a recession signal. View Insider Data button available.

4. Yield Rate of Change Chart

Recharts LineChart. Two lines: 2yr yield RoC (blue) and 10yr yield RoC (orange). Three toggle buttons: Levels (actual % values) / Spread (differential) / RoC (daily % change). Period selector: 1W / 1M / 3M. Zero line prominently marked on RoC view. Green shading above zero, red below. Hover tooltip shows exact values at any date point.

5. Economic Data Panel

Four cards side by side: CPI (inflation), GDP (growth), PMI (business activity), Employment. Each card: latest reading, arrow showing change vs previous, previous reading, AI label (Cooling/Growing/Expanding/Contracting/Stable), next release date. View Insidr Data button on each card.

6. Country Risk Sentiment

Current Risk On/Off label and its specific implication for that country's currency. Pulled from the existing Risk On/Off engine — no separate calculation needed. View Insidr Data button available.

7. Recent High Impact News

Last 5 high-impact news articles tagged to this country's currency. Simplified card format: impact badge, sentiment badge, headline, timestamp. View Insidr Data button on each article.

Country Indicator Mapping

Country	CPI	GDP	PMI	Employment
USA	CPI YoY	GDP QoQ	ISM Manufacturing PMI	Non-Farm Payrolls
Eurozone	HICP YoY	Eurozone GDP QoQ	Eurozone Manufacturing PMI	Eurozone Unemployment
UK	CPI YoY	GDP QoQ	UK Manufacturing PMI	Claimant Count
Japan	CPI YoY	GDP QoQ	Jibun Bank PMI	Unemployment Rate
Australia	CPI YoY	GDP QoQ	Judo Bank PMI	Employment Change
New Zealand	CPI YoY	GDP QoQ	BusinessNZ PMI	Employment Change
Canada	CPI YoY	GDP QoQ	Ivey PMI	Employment Change
Switzerland	CPI YoY	GDP QoQ	procure.ch PMI	Unemployment Rate

Country	CPI	GDP	PMI	Employment
China	CPI YoY	GDP YoY	Caixin Manufacturing PMI	Unemployment Rate

SECTION 10: YIELD DIFFERENTIAL ENGINE

Yield Differential Engine

What It Tracks

2-year and 10-year government bond yields for 8 countries: USA, Germany, UK, Japan, Australia, New Zealand, Canada, Switzerland. Updated every 15 minutes.

Instruments Tracked

Country	2yr Instrument	10yr Instrument
USA	US2Y	US10Y
Germany (EUR proxy)	DE2Y	DE10Y (Bund)
United Kingdom	GB2Y	GB10Y (Gilt)
Japan	JP2Y	JP10Y (JGB)
Australia	AU2Y	AU10Y
New Zealand	NZ2Y	NZ10Y
Canada	CA2Y	CA10Y
Switzerland	CH2Y	CH10Y

What Gets Calculated Per Pair

- 2yr spread — Country A 2yr yield minus Country B 2yr yield
- 10yr spread — Country A 10yr yield minus Country B 10yr yield
- Spread direction — widening or narrowing, and in which country's favour
- Pair alignment — whether price movement is aligned with or diverging from yield direction
- Divergence flag — when price is moving opposite to what yields suggest, flagged as potential mean reversion opportunity
- Daily rate of change — percentage change in each yield instrument day over day

Pair to Yield Mapping

Pair	Country A Yields	Country B Yields
EUR/USD	German 2yr + 10yr	US 2yr + 10yr
GBP/USD	UK 2yr + 10yr	US 2yr + 10yr
USD/JPY	US 2yr + 10yr	Japan 2yr + 10yr
AUD/USD	Australian 2yr + 10yr	US 2yr + 10yr
NZD/USD	NZ 2yr + 10yr	US 2yr + 10yr
USD/CAD	US 2yr + 10yr	Canadian 2yr + 10yr
USD/CHF	US 2yr + 10yr	Swiss 2yr + 10yr
EUR/GBP	German 2yr + 10yr	UK 2yr + 10yr
GBP/JPY	UK 2yr + 10yr	Japan 2yr + 10yr
EUR/JPY	German 2yr + 10yr	Japan 2yr + 10yr
AUD/JPY	Australian 2yr + 10yr	Japan 2yr + 10yr
XAU/USD	US 10yr real yield	DXY as proxy

Trade Idea Confluence Integration

Yield Condition	Confluence Points
Both 2yr and 10yr spreads confirm idea direction	+3
2yr spread only confirms direction	+2
10yr spread only confirms direction	+1
Spreads neutral or flat	0
2yr spread contradicts direction	-2
Both spreads contradict direction	-3
Price/yield divergence detected (mean reversion opportunity)	+2

SECTION 11: RISK ON / RISK OFF ENGINE

Risk On / Risk Off Engine

Global macro sentiment scored every 15 minutes across seven instruments. Outputs a three-state label used to filter trade ideas and inform users of the macro environment.

Seven Indicators

Indicator	Risk Off Signal	Neutral	Risk On Signal
VIX	Above 20 and rising	15 to 20	Below 15 and falling
S&P 500	Down more than 0.5% on day	Flat +/- 0.5%	Up more than 0.5% on day
JPY	Strengthening (USD/JPY falling)	Ranging	Weakening (USD/JPY rising)
Gold	Rising more than 0.3%	Flat	Falling more than 0.3%
US 10Y Yield	Falling — flight to bonds	Flat	Rising — risk appetite
DXY	Surging — USD safe haven bid	Flat	Falling — risk on
Commodity Index	Falling — demand concerns	Flat	Rising — growth optimism

Scoring Logic

Each indicator votes +1 (Risk On), 0 (Neutral), or -1 (Risk Off). Sum range: -7 to +7.

Score Range	Label
+3 to +7	Risk On
-2 to +2	Neutral
-3 to -7	Risk Off

Asset Alignment Rules

Environment	Favours Long	Favours Short
Risk On	AUD/USD, NZD/USD, AUD/JPY, NZD/JPY, GBP/JPY, EUR/JPY, Equities, Commodities	Gold, USD/CHF, Defensive assets
Risk Off	Gold (XAU/USD), USD/JPY, USD/CAD	AUD/USD, NZD/USD, AUD/JPY, Equities, Commodities
Neutral	No boost or suppression applied	Technical signals carry full weight

Platform Usage

- Dashboard — Risk On/Off widget showing current label and all 7 indicator votes
- Trade Ideas — +2 confluence points when idea aligns, -3 when it contradicts
- Every trade idea card — alignment badge shown
- Risk Sentiment Shift Alerts — users notified when label changes state
- When label changes state: all active trade ideas are re-evaluated against the new environment

SECTION 12: WATCHLIST

Watchlist

The Watchlist is the user's personal asset monitor. It does not have its own detail page — clicking any asset loads it on the Dashboard as the selected asset breakdown.

Display

- Assets grouped by market: Forex / Stocks / Indices / Crypto
- Each market group is collapsible
- Each group header: market emoji + name + asset count + Edit button
- Each asset row: symbol (JetBrains Mono bold), full name on hover, live price updating via WebSocket (green flash on uptick, red flash on downtick), 24hr percentage change with arrow, AI bias badge (Bullish/Bearish/Neutral with strength)
- Hover on bias badge shows tooltip with Swing bias, Day bias, and last updated time
- Row hover actions: quick alert shortcut, remove from watchlist, more options
- Empty group shows prompt to add assets

Add Asset

- Plus button opens search modal
- Search input with real-time results: asset symbol + full name + market type badge
- Selecting an asset adds it to the correct market group automatically
- Powered by Twelve Data symbol search endpoint

Clicking an Asset

Navigates to /dashboard and loads that asset as the selected pair. The pairs tab strip updates to show the selected asset. All dashboard asset breakdown content updates to that asset.

SECTION 13: TRADE JOURNAL

Trade Journal & Portfolio Tracker

A comprehensive Tradezella-style trading journal and performance analytics system. Once users log trades and build performance history they are deeply invested in the platform — this is a primary retention driver.

Four Tabs

- Overview — performance summary, recent trades, key metrics
- Journal — full trade log with filters and entry form
- Analytics — deep performance breakdowns and charts
- Calendar — monthly calendar view with P&L per day

What Users Log Per Trade

Field	Type	Notes
Asset	Searchable select	Any symbol across all 4 markets
Market	Select	Forex / Stock / Index / Crypto
Direction	Select	Long or Short
Entry Price	Number	
Exit Price	Number	
Stop Loss	Number	
Take Profit	Number	
Position Size	Number	Lots, units, shares, or dollar amount
Entry Date/Time	DateTime	Stored UTC, displayed in user timezone
Exit Date/Time	DateTime	
Timeframe	Select	1m, 5m, 15m, 1H, 4H, Daily
Strategy	Select/Text	User-defined strategies
Emotional State	Select	Calm, Anxious, Overconfident, FOMO

Field	Type	Notes
Mistake Tags	Multi-select	Pre-built list plus custom tags
Notes	Free text	Open journal entry
Screenshot	Image upload	Stored in Supabase Storage
P&L	Auto-calculated	In dollar amount and R multiples

Pre-Built Mistake Tags

Moved stop loss, Overtraded, Revenge trade, Entered too early, Entered too late, Ignored trading plan, Traded during news, Poor risk management, FOMO entry. Users can also create custom tags.

Trade Import Options

- Manual entry — form-based with all fields above
- CSV import — pre-built parsers for MT4/MT5, IBKR, Tradovate, Binance, Coinbase, Robinhood. Custom CSV mapper for unsupported brokers.
- MT4/MT5 automatic sync — custom Expert Advisor (EA) built in MQL4/MQL5 that pushes executed trades to the platform backend API in real time. Critical integration as the majority of retail forex traders use MetaTrader.
- IBKR Flex Web Service — automated trade history sync for IBKR users
- Binance API — crypto trade history sync
- Coinbase Advanced API — crypto trade history sync

Analytics Breakdowns

Performance is broken down by: market (Forex/Stocks/Indices/Crypto), trading session (London/NY/Tokyo/Sydney), day of week, timeframe, strategy, emotional state, and mistake tag. Each breakdown shows trade count, win rate, average P&L, and total P&L.

Charts and Visualisations

- Equity curve — cumulative P&L over time (Recharts LineChart)
- Monthly P&L bar chart (Recharts BarChart)
- Win/loss distribution (Recharts PieChart)
- Drawdown chart
- Mistake cost tracker — each tag with total count and total dollar cost

Calendar View

- Monthly calendar grid
- Each day: green background for profitable, red for losing, grey for no trades
- P&L amount displayed in each day cell
- Clicking any day expands to show all trades taken that session
- Monthly summary stats at top of calendar

SECTION 14: ALERTS & NOTIFICATIONS

Alerts & Notifications

Five Alert Types

Alert Type	Triggers When	Configurable Filters
Trade Idea Alert	New idea published for a watched asset	Confidence level (80%+ only etc), direction (Long/Short/Any), timeframe type (Scalp/Day Trade/Swing/Any)
Economic Calendar Alert	X minutes before a scheduled event releases for currencies in user's watchlist	Lead time (15min/30min/1hr/2hr), impact level (High only / High + Medium), watchlist filter
News Alert	High or medium impact article published for a watched asset	Impact level, sentiment direction (Bullish only / Bearish only / Any)
Risk Sentiment Shift Alert	Global Risk On/Off environment changes state	Direction (Any shift / Only Risk Off / Only Risk On)
Bias Change Alert	AI bias flips direction for a watched asset	Timeframe (Swing / Day / Both), flip direction (Any / Bullish to Bearish / Bearish to Bullish)

Delivery Channels — Launch Version

Two channels at launch. SMS to be added in a future release.

Channel	Provider	Limit	Notes
In-App	PostgreSQL (Supabase)	Unlimited	Bell icon in top bar, 30-day notification history
Email	SendGrid or Resend	Unlimited	Transactional email for all alert types

Notification Preferences Matrix

Every alert type has independent on/off toggles for each channel. Users configure this in Settings > Notifications. Quiet hours setting suppresses all notifications between user-configured times.

Alert Type	In-App Toggle	Email Toggle
Trade Idea Alerts	On/Off	On/Off
Economic Calendar Alerts	On/Off	On/Off
News Alerts	On/Off	On/Off
Risk Sentiment Shift Alerts	On/Off	On/Off
Bias Change Alerts	On/Off	On/Off

Alert Management Page

- Alerts grouped by type with count badge per group
- Filter tabs: All / Active / Triggered Today / Paused
- Each alert card: type icon, asset (if applicable), configuration summary, channel icons, Edit/Pause/Delete actions
- Create Alert button — 4 step modal: type selection, configuration, channel selection, confirmation
- Economic Calendar events and individual news articles also have Set Alert shortcut buttons inline

In-App Notification Centre

- Bell icon in top bar with unread count badge
- Click opens dropdown showing last 20 notifications
- Each notification: type icon, title, description, time ago, action deep link, read/unread state
- Mark all read button + View All link to /alerts page
- Notifications stored for 30 days then auto-archived

SECTION 15: AI CHAT ASSISTANT

AI Chat Assistant

A persistent AI trading mentor available on every authenticated page via a floating widget in the bottom-right corner. It knows each user's trading history, watchlist, and current market conditions — it is not a generic chatbot.

Widget Behaviour

- Fixed bottom-right corner on every authenticated page
- Collapsed: circular button (48px) with purple-to-gold gradient and star icon. Pulse animation on new market events.
- Expanded: slides up into chat panel (400px wide, 600px tall) with smooth animation
- Persists across page navigation — conversation does not reset when user changes page
- Header: AI Assistant title + gradient accent + minimize button

Capabilities

- Answer general trading questions in plain English
- Analyse any specific asset on demand using live price, current bias, VWAP, 200 EMA, and recent news
- Review user's past trades from the journal and identify specific weaknesses
- Explain economic events and news articles with market impact context
- Educate on trading concepts calibrated to the user's experience level (Beginner/Intermediate/Advanced set during onboarding)

Persistent Memory

- Full conversation history stored in PostgreSQL
- Last 20 messages loaded as context on each new message
- User profile summary injected into every system prompt: name, preferred markets, experience level, win rate, most common mistake tags, last 5 trades
- Context refreshed daily from the user's journal data

Rate Limiting

- Pro tier: 50 messages per day
- Elite tier: unlimited

Guardrails

- Never says you should buy or sell a specific asset
- Always frames responses as analysis and education, not advice
- Every trade-related response ends with: This is not financial advice — always do your own research

Starter Prompts

Four suggested prompts shown when chat first opens or is empty:

- What is the market looking like today?
- Analyse EUR/USD for me
- Review my trading this week
- Explain the latest Fed decision

SECTION 16: VIEW INSIDR DATA

View Insidr Data — Universal Button

View Insidr Data is a contextual magnifying glass icon button that appears next to every individual item across the platform. It generates a focused Claude-powered AI briefing about that specific item only. It is never a page-level summary — always item-level intelligence.

Where It Appears

Location	Item Level	What It Generates
Dashboard — asset breakdown	Per selected asset	Complete briefing: bias, yield direction and RoC, Risk On/Off alignment, key price levels, news sentiment, fundamental driver context, upcoming events
News feed — every article	Per article	What this specific article means for the affected assets, which direction it pushes them, and what to watch
Economic Calendar — every event	Per event	What this event measures, the three scenarios (above/below/inline forecast), which assets are most affected and how
Trade Ideas — every idea card	Per idea	Why this specific setup was identified, which confluence signals fired, what to watch for
Economy pages — Central Bank Panel	Per CB panel	Current central bank stance and its implications for that currency
Economy pages — Economic Data Panel	Per indicator card	What this specific reading means for the economy and currency
Economy pages — News articles	Per article	Same as news feed treatment
Watchlist — bias badge	Per asset	What is driving the current bias and what levels to watch

Technical Implementation

- Small magnifying glass icon — consistent purple accent colour across all instances
- On click: expands inline panel below the item or opens a clean modal depending on context
- Generated by Claude API (claude-sonnet-4-20250514) on demand

- Data fed to Claude comes entirely from existing cached platform data — no new API calls needed for data collection
- Cached per item for 15-30 minutes in Redis — if the same item is clicked again within the cache window, the cached response is served instantly
- All platform data (bias cache, yield cache, risk engine, news pipeline) is already collected for other features — View Insider Data simply aggregates what is already known and asks Claude to synthesise it

SECTION 17: SETTINGS & SUBSCRIPTION

Settings & Subscription

Settings Navigation

Left sub-navigation within the Settings page with four sections: Profile, Notifications, Markets and Display, Subscription and Billing.

Profile

- First name, last name, email with verified badge, password change
- Avatar upload
- Experience level select (Beginner/Intermediate/Advanced) — affects AI chat tone
- Preferred name — what the AI assistant calls the user
- Delete Account — buried at the very bottom of the page as a small understated text link only. Requires a warning modal then user must type DELETE to confirm. Account deleted immediately upon confirmation with no grace period.

Notifications

- Matrix grid: rows = 5 alert types, columns = 2 channels (In-app / Email)
- Toggle per cell — each alert type independently controllable per channel
- Quiet hours: enable toggle + from/to time picker. No notifications sent during quiet window.

Markets and Display

- Preferred markets toggles: Forex / Stocks / Indices / Crypto — affects what the platform surfaces by default
- Default dashboard asset — which asset loads on the chart when Dashboard opens
- Theme: Dark only — toggle is present but disabled (always dark)
- Chart default timeframe: 15m / 1H / 4H / Daily
- Display currency: USD / GBP / EUR / AUD
- Timezone: auto-detect from browser toggle + manual timezone select
- Number format: regional select for decimal and thousands separator

Subscription and Billing

- Current plan banner: plan name, price, next billing date, status badge (Active/Cancelled/Past Due)
- Plan comparison table: Pro vs Elite with full feature list
- Monthly/Annual billing toggle — annual saves 2 months
- Current payment method: card last 4 digits + expiry + Update button (opens Stripe hosted payment update)
- Upgrade/downgrade: upgrades are immediate with Stripe proration, downgrades take effect at end of current billing period
- Cancellation flow: small text link (not a button) > warning modal explaining data retention > user types DELETE > account cancelled immediately, access continues to end of billing period

Subscription Tiers

Feature	Pro — \$28/mo	Elite — \$79/mo
Trade Ideas	Unlimited	Unlimited + Priority
AI Chat	50 messages/day	Unlimited
News Feed	Full access	Full access
Economic Calendar + AI	Full access	Full access
Economy Pages	Full access	Full access
Watchlist	Full access	Full access
Trade Journal	Full access	Full access
Alerts	In-app + Email	In-app + Email + SMS (future)
View Insider Data	Full access	Full access
Annual Billing	\$280/yr (save \$56)	\$790/yr (save \$158)

SECTION 18: SUPABASE DATABASE SCHEMA

Supabase Database Schema

All tables use Row Level Security (RLS). Users can only read and write their own data. Enable RLS on every table from day one.

User Tables

Table	Key Fields
users	id, email, first_name, last_name, avatar_url, preferred_name, experience_level, created_at
user_preferences	id, user_id, timezone, theme, chart_default_timeframe, preferred_markets (jsonb), default_dashboard_asset, display_currency, quiet_hours_enabled, quiet_hours_from, quiet_hours_to
subscriptions	id, user_id, plan (pro/elite), billing_cycle (monthly/annual), status (active/cancelled/past_due), stripe_subscription_id, stripe_customer_id, current_period_end, cancel_at_period_end
onboarding	id, user_id, completed_at, steps_completed (jsonb), plan_selected, trial_started

Market Data Cache Tables

Table	Key Fields
user_watchlist	id, user_id, asset, market, display_order, added_at
asset_bias_cache	id, asset, swing_bias, swing_strength, day_bias, day_strength, news_impact_level, supports_bias (jsonb), flips_bias (jsonb), generated_at
asset_levels_cache	id, asset, resistance_levels (jsonb), support_levels (jsonb), poc_level, prev_day_high, prev_day_low, prev_week_high, prev_week_low, prev_month_high, prev_month_low, session_vwap, weekly_vwap, ema_200_4h, ema_200_daily, calculated_at
risk_sentiment_cache	id, label (risk_on/risk_off/neutral), total_score, vix_vote, sp500_vote, jpy_vote, gold_vote, yield_vote, dxy_vote, commodity_vote, calculated_at

Table	Key Fields
yield_data	id, instrument, country, maturity (2yr/10yr), yield_value, change_bps, fetched_at
pair_yield_spreads	id, asset, spread_2yr, spread_10yr, favours, pair_aligned (boolean), divergence_detected (boolean), ai_analysis, calculated_at
yield_roc_history	id, instrument, country, maturity, yield_value, roc_daily (%), recorded_date — retained 1 year
country_economy_cache	id, country, cb_current_rate, cb_next_meeting_date, cb_rate_expectations (jsonb), cpi_latest, cpi_previous, cpi_next_release, gdp_latest, gdp_previous, pmi_latest, pmi_previous, employment_latest, employment_previous, ai_macro_summary, calculated_at

Trade Ideas Tables

Table	Key Fields
trade_ideas	id, asset, market, direction (long/short), timeframe_type (scalp/day_trade/swing), chart_timeframes (jsonb), confidence (0-100), entry_low, entry_high, stop_loss, tp1, tp2, tp3, rr_tp1, rr_tp2, rr_tp3, invalidation_level, invalidation_description, technical_reasoning, fundamental_reasoning, risk_env_label, risk_env_aligned (boolean), status, generated_at, expires_at
idea_status_history	id, idea_id, status, price_at_update, updated_at
platform_performance_stats	id, period (daily/monthly), total_ideas, wins, losses, win_rate, avg_rr, tp1_hit_rate, tp2_hit_rate, tp3_hit_rate, calculated_at
scanner_results	id, asset, timeframe, confluence_score, signals_triggered (jsonb), fundamental_score, risk_env_score, yield_score, idea_generated (boolean), scanned_at

News and Calendar Tables

Table	Key Fields
news_articles	id, headline, summary, full_content, source, source_url, published_at, fetched_at, market, impact_level, sentiment, sentiment_score (-1 to 1), asset_tags (jsonb), ai_analysis
market_sentiment	id, market, score (0-100), label, driver_text, news_score, momentum_score, volatility_score, calculated_at
asset_sentiment_scores	id, asset, market, sentiment_score, sentiment_label, price_change_24h, calculated_at
economic_events	id, event_name, currency, country, impact_level, event_datetime (UTC), previous_value, forecast_value, actual_value, beat_miss_inline

Table	Key Fields
	(beat/miss/inline/pending), affected_assets (jsonb), ai_explanation, ai_impact_prediction, ai_post_analysis, post_analysis_at

Journal Tables

Table	Key Fields
trades	id, user_id, asset, market, direction, entry_price, exit_price, stop_loss, take_profit, position_size, entry_time, exit_time, timeframe, strategy, emotional_state, notes, screenshot_url, pnl, pnl_r, result (win/loss/breakeven), created_at
mistake_tags	id, trade_id, tag_name
user_strategies	id, user_id, strategy_name, description
broker_connections	id, user_id, broker_name, api_key_encrypted, last_sync_at

Alerts and Chat Tables

Table	Key Fields
user_alerts	id, user_id, alert_type (trade_idea/calendar/news/risk/bias), asset, is_active, is_paused, config (jsonb), channels (jsonb: inapp/email), created_at, last_triggered_at
notifications	id, user_id, alert_id, type, title, body, action_url, is_read, delivered_inapp, delivered_email, created_at, read_at — 30 day retention
chat_conversations	id, user_id, title (auto-generated from first message), created_at, last_message_at
chat_messages	id, conversation_id, user_id, role (user/assistant), content, created_at
user_ai_profile	id, user_id, experience_level, preferred_markets (jsonb), performance_summary (jsonb — cached daily from journal), last_updated

SECTION 19: RECOMMENDED TECHNOLOGY STACK

Recommended Technology Stack

Advisory Note

The following technology stack is a recommendation based on the platform's requirements. Final stack decisions rest with the development team. Each recommendation includes the rationale specific to The Insider's needs and notes alternatives where relevant.

Frontend

Technology	Purpose	Rationale
React	UI framework	Component-based, industry standard, large ecosystem, works well with all other recommended tools
Tailwind CSS	Styling	Utility-first, fast to build with, easy to enforce the design system through CSS variables
Recharts	Data charts	Yield RoC charts, journal equity curve, analytics visualisations — React-native, no external dependencies
TradingView Widget	Candlestick charts	Free embed, industry standard, handles all chart rendering internally — no alternative needed
React Query (TanStack)	Server state	Caching, background refetch, loading states for all API-dependent content
Zustand	Global state	Lightweight, simple — for user preferences, watchlist state, selected asset

Backend

Technology	Purpose	Rationale
Python + FastAPI	API server	Async, fast, excellent for AI and data workloads, auto-generates API docs
APScheduler or Celery	Background jobs	Trade idea scanner (every 15 min), bias refresh (every 30 min), yield updates (every 15 min), news ingestion (every 60 sec)
Redis	Caching layer	All short-TTL data: bias cache, sentiment scores, yield

Technology	Purpose	Rationale
		cache, risk engine state, View Insider Data responses
Anthropic Python SDK	Claude API	All AI features — trade ideas, bias, chat, sentiment, View Insider Data

Database and Auth

Technology	Purpose	Rationale
Supabase (PostgreSQL)	Primary database	Handles database, authentication, file storage, and real-time subscriptions in one service
Supabase Auth	Authentication	Email/password, JWT sessions, password reset — no separate auth service needed
Supabase Storage	File storage	Trade screenshot images from the journal

External Services

Service	Purpose	Notes
Stripe	Subscription billing	Handles Pro and Elite tiers, monthly/annual billing, proration, webhooks, customer portal
SendGrid or Resend	Email	Transactional email for alerts, auth emails, billing receipts
PostHog or Mixpanel	Analytics	Onboarding funnel tracking, user behaviour, feature usage
Vercel	Frontend hosting	CI/CD, CDN, preview deployments from GitHub
Railway	Backend hosting	FastAPI server, background job workers, Redis instance

Market Data APIs

API	Purpose	Est. Monthly Cost
Twelve Data	Live prices (WebSocket), OHLCV, indicators, economic calendar, yield instruments, symbol search	\$30-60
Trading Economics	Central bank rates, rate expectations, NZ and CH yields, Chinese macro data	\$50
Benzinga API	Financial news — stocks, forex, indices	\$50
CryptoPanic API	Crypto-specific news	Free tier
Anthropic Claude API	All AI features — claude-sonnet-4-20250514	\$50-200 (usage-based)

SECTION 20: ANALYTICS & TRACKING

Analytics & Funnel Tracking

Analytics tracking serves two purposes: understanding user behaviour to improve the product, and monitoring the health of the business. Recommended tool: PostHog (self-hostable, privacy-friendly) or Mixpanel.

Onboarding Funnel Events

Event	Properties
onboarding_step_viewed	step_name, step_number, user_id, timestamp
onboarding_step_completed	step_name, step_number, time_on_step (seconds), user_id
onboarding_step_back	step_name, user_id
onboarding_abandoned	step_name, step_number, time_in_flow (seconds), user_id
onboarding_completed	total_time (seconds), markets_selected, experience_level, user_id
plan_selected	plan (pro/elite), billing_cycle (monthly/annual), user_id
trial_started	plan, billing_cycle, user_id

Core Product Events

Event	Properties
trade_idea_viewed	idea_id, asset, direction, timeframe_type, confidence, user_id
trade_idea_alert_set	idea_id, asset, user_id
trade_idea_added_to_journal	idea_id, asset, user_id
view_insidr_data_clicked	item_type, item_id, asset, page, user_id
asset_selected	asset, source (tab_strip/watchlist), user_id
yield_chart_toggle_changed	view (levels/spread/roc), asset, user_id
news_article_expanded	article_id, impact_level, sentiment, user_id
alert_created	alert_type, channels, user_id
trade_logged	market, direction, result, user_id

Event	Properties
chat_message_sent	message_count_today, user_id
subscription_upgraded	from_plan, to_plan, user_id
subscription_cancelled	plan, reason, user_id

SECTION 21: MACHINE LEARNING ROADMAP

Machine Learning Roadmap

Launch Requirement

None of the V1 platform features require training a machine learning model. The platform launches entirely on Claude API with prompt engineering. ML features are a future roadmap item that improve the platform over time as outcome data accumulates.

V1 — Available at Launch

FinBERT Financial Sentiment Scoring

FinBERT is a pre-trained NLP model fine-tuned on financial text. It classifies news articles as bullish, bearish, or neutral significantly faster and cheaper than Claude API. Available immediately from HuggingFace (ProsusAI/finbert). Recommended to deploy from day one to reduce Claude API costs on news sentiment scoring. Install: `pip install transformers torch`.

V2 — 3 to 6 Months Post-Launch

Trade Idea Quality Scoring (XGBoost)

Once the platform has published approximately 500 to 1,000 ideas with recorded outcomes, an XGBoost model can be trained to predict the probability of a successful outcome for any new idea based on its full feature set. The confidence score transitions from a rule-based confluence count to an evidence-based probability derived from actual historical performance. Requires: `scikit-learn`, `xgboost`.

Personalised Idea Ranking

A recommendation model that learns each user's preferences from their behaviour — which ideas they click, save, set alerts for, and log to their journal. Re-ranks the ideas feed per user. Requires 200+ active users with sufficient interaction data.

V3 — 12+ Months Post-Launch

Price Movement Probability Model

A model trained on historical OHLCV data that predicts the probability distribution of price movement over the next N candles for a given market condition. Adds a statistical probability layer alongside the existing confidence score.

Computer Vision Chart Pattern Detection

A CNN trained on labelled chart images to detect patterns directly from visual price representation. More accurate than algorithmic detection. Requires labelled training data and GPU infrastructure for initial training.

User Churn Prediction

A model trained on user behaviour data to predict which subscribers are at risk of cancelling. Triggers automated retention actions when a user crosses a risk threshold. Requires 6+ months of user data with sufficient churned users to train on.

Market Anomaly Detection

An Isolation Forest or Autoencoder model trained on normal multi-market behaviour. Flags when multiple markets simultaneously exhibit unusual correlated behaviour — providing earlier warning of risk events than the rule-based Risk On/Off system. Requires 6+ months of platform market data.

SECTION 22: LEGAL & REGULATORY

Legal & Regulatory Requirements

Important

The development team should seek independent legal counsel in each jurisdiction of operation to confirm that the platform's activities fall within financial information provision rather than regulated financial advice. Regulations vary significantly between Australia, the UK, the EU, and the US.

Financial Information Disclaimer

The platform provides AI-generated trade ideas and market analysis for informational and educational purposes only. It does not execute trades, manage funds, or provide personalised financial advice. This disclaimer must appear on:

- The landing page (prominent placement)
- Every trade idea card and detail page
- The AI Chat Assistant (end of every trade-related response)
- The Terms of Service
- The onboarding flow

Required Legal Documents

- Terms of Service — covering platform use, subscription terms, cancellation policy, refund policy, limitation of liability
- Privacy Policy — GDPR compliant for EU users, covers data collection, retention, and deletion
- Financial Information Disclaimer — clearly stating the platform does not provide financial advice
- Cookie Policy

Data Privacy

- GDPR compliance required for EU users
- Delete account must fully purge all user data from the system

- Data retention policies must be clearly stated in the Privacy Policy
- Supabase handles data storage — ensure data residency requirements are met for target markets

Payment and Billing

- Stripe handles all payment processing and is PCI-DSS compliant — the platform never touches raw card data
- Subscription terms, cancellation policy, and refund policy must be clearly stated
- 7-day free trial terms must be clearly communicated before Stripe Checkout

Platform Performance Transparency

The publicly displayed platform performance statistics (win rate, TP hit rates) must include a disclaimer stating that past performance of AI-generated trade ideas does not guarantee future results and that statistics are based on the platform's own tracking methodology.